THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a v-t graph, a line sloping downward from 20 m/s to 0 in 4 s shows retardation of: 1

(a) 5 m/s squared **(b)** 80 m/s squared **(c)** 4 m/s squared **(d)** 20 m/s squared

2. The distance covered by a body after half a circle of radius r is: 1

(a) zero **(b)** 2 pi r **(c)** 2r **(d)** pi r

3. A body starting from rest with a = 2 m/s squared covers 25 m. Its final velocity is: 1

(a) 10 m/s **(b)** 50 m/s **(c)** 5 m/s **(d)** 20 m/s

4. A car brakes uniformly from 20 m/s to rest in 4 s. The area under its v-t graph, which is the distance covered, is: 1

(a) 80 m **(b)** 5 m **(c)** 20 m **(d)** 40 m

5. A car with a speed of 54 km/h is 150 m from a signal. If it keeps the speed, the time to reach the signal is: 1

(a) 2.8 s **(b)** 5 s **(c)** 15 s **(d)** 10 s

6. A stone is dropped; after 3 s it has fallen (g = 10): 1

(a) 90 m **(b)** 45 m **(c)** 30 m **(d)** 15 m

7. A car moves with uniform acceleration. The distance covered in the nth second is proportional to: 1

(a) n squared **(b)** 1/n **(c)** n **(d)** (2n - 1)

8. A train 200 m long passing a pole in 10 s has speed: 1

(a) 10 m/s **(b)** 200 m/s **(c)** 2 m/s **(d)** 20 m/s

9. The velocity of a stone dropped for 3 s is (g = 10): 1

(a) 10 m/s **(b)** 30 m/s **(c)** 15 m/s **(d)** 90 m/s

10. The distances covered in successive seconds by a freely falling body from rest are in the ratio: 1

(a) 1 : 3 : 5 **(b)** 1 : 1 : 1 **(c)** 1 : 4 : 9 **(d)** 1 : 2 : 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State why the slope of an s-t graph gives speed and show with numbers: 20 m at 2 s and 60 m at 6 s. 2

12. An athlete runs 100 m in 10 s. Find the average speed in m/s and km/h. 2

13. A body of velocity 10 m/s accelerates at 2 m/s squared for 5 s. Find the final velocity and the distance covered. 2

14. State the average velocity in a round trip and explain why. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The speed of a car increases from 36 km/h to 72 km/h in 10 s. Find the acceleration in m/s squared and the distance travelled. 3

16. Explain the meaning of a v-t graph with equal horizontal segments at 5 m/s, then a rising line, then horizontal at 10 m/s. 3

17. A car moves 30 km in the first hour and 50 km in the second hour in the same direction. Find the average speed and say if the motion is uniform. 3

OR

A body starts from rest and moves with a = 4 m/s squared. Find the distance in the 3rd second. 3

18. A car travels with speeds of 20 m/s for 10 s, 30 m/s for 10 s and 40 m/s for 10 s. Find the average speed. 3

OR

A ball rolls down an incline from rest with an acceleration of 0.5 m/s squared. Find the velocity after 6 s and the distance covered. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how to find the displacement, the acceleration and the average velocity from a v-t graph given as: 0 to 4 s rising from 0 to 8 m/s; 4 to 8 s constant at 8 m/s; 8 to 10 s falling to 0. 5

OR

The table gives t(s): 0, 1, 2, 3, 4 and s(m): 0, 5, 20, 45, 80 for a body. Analyse the motion, find the acceleration and the velocity at t = 4 s. 5

20. Two stations A and B are 120 km apart. A train leaves A at 60 km/h towards B and another leaves B at 40 km/h towards A at the same time. After how long and where do they meet? 5

OR

Explain, using the relative speed idea, why a passenger in a train moving at 80 km/h sees an oncoming train of speed 70 km/h passing at 150 km/h. Find the time taken to cross a 300 m train. 5

END OF QUESTION PAPER — Total: 40 Marks